Home Assignment Week-03- Day 2

Task : With vibe coding implement the below

1) Connect to jira via api
2) Fetch the user storie from jira to generate testcases

Step 1: JIRA Integration
Prompt: Analyze the existing codebase and implement JIRA integration. **GO STEP BY STEP — upon my approval, proceed to the next step.**
[MANDATORY] Backend first, then Frontend. –
[CRITICAL] Email and API key must be configurable. –
[CRITICAL] Projects must be fetched from JIRA and auto-populated as a dropdown. --
[CRITICAL] User stories for the selected project must be fetched and auto-populated as a dropdown.
-- When a story is selected, fetch its details and **generate test cases** when **Generate** is clicked.
-- When a story is selected, **do not** require manual input — fields auto-populate and are ready to generate.
-- Look for acceptance criteria in acceptanceCriteria; if empty, search the **Description** field after the keyword **"Acceptance Criteria"** and use that content as acceptance criteria in the UI. **Reference:** Postman collection and provided JIRA instance.

CURRENT CODEBASE ANALYSIS (summary)
Architecture:

- **Backend:** Express.js + Groq LLM integration
- **Frontend:** React + Vite
- **Current Flow:** Manual input → LLM generates test cases **To Add:** Jira integration with dynamic project/story selection

STEP-BY-STEP IMPLEMENTATION PLAN
PHASE 1 — BACKEND: JIRA CONNECTIVITY & CONFIGURATION

1. **Install Jira client library** (e.g., jira-client or @atlassian/jira).
2. **Create Jira configuration schema** (backend/src/schemas.ts) — add types for Jira config, Project, Story, StoryDetails.
3. **Update environment variables** (root .env) — make **JIRA_EMAIL**, **JIRA_API_KEY**, **JIRA_HOST** configurable.
4. **Create Jira client** (backend/src/llm/jiraClient.ts):
    - Initialize client with auth (email + API key).
    - Implement methods:
        - getProjects() — fetch all projects.
        - getStories(projectKey) — fetch user stories for a project.

        - getStoryDetails(storyKey) — fetch full story including acceptance criteria. **Approval:** Proceed with Phase 1 backend setup? ✓ / ✗

PHASE 2 — BACKEND: NEW API ENDPOINTS

Create routes (backend/src/routes/jira.ts):

- POST /api/jira/config — Save and validate Jira credentials.
- GET /api/jira/projects — Return list of projects.
- GET /api/jira/stories?projectKey=XXX — Return stories for a project.
- POST /api/generate-tests — Accept storyKey (instead of manual input) and trigger test generation. **Approval:** Proceed with Phase 2? ✓ / ✗

PHASE 3 — FRONTEND: TYPES & API SETUP

1. **Update** types.ts with Jira types (Config, Project, Story, StoryDetails).
2. **Update** api.ts with functions:
    - saveJiraConfig(config) — POST credentials to backend.
    - fetchProjects() — GET projects.
    - fetchStories(projectKey) — GET stories.
    - generateTestsFromStory(storyKey) — POST storyKey to generate tests. **Approval:** Proceed with Phase 3? ✓ / ✗

PHASE 4 — FRONTEND: UI COMPONENTS

1. **Modify** App.tsx**:**
    - Add Jira credentials form (email, API key, host).
    - Replace manual story input with:
        - **Dropdown 1:** Select Project (auto-populated).
        - **Dropdown 2:** Select User Story (auto-populated after project selection).
        - **Button:** Generate Tests (calls backend with story key).
2. **Create** frontend/src/components/JiraConfig.tsx**:** Form to save/validate Jira credentials. **Approval:** Implement Phase 4 UI components? ✓ / ✗

PRE-APPROVAL CHECKLIST (before starting)

Please confirm/provide:

- Postman collection reference to use for API details.
- JIRA instance URL and default project to test with.
- API authentication preference: **API Token** (preferred) or OAuth.
- Any default project/stories to validate against.

Provided references (from your message):

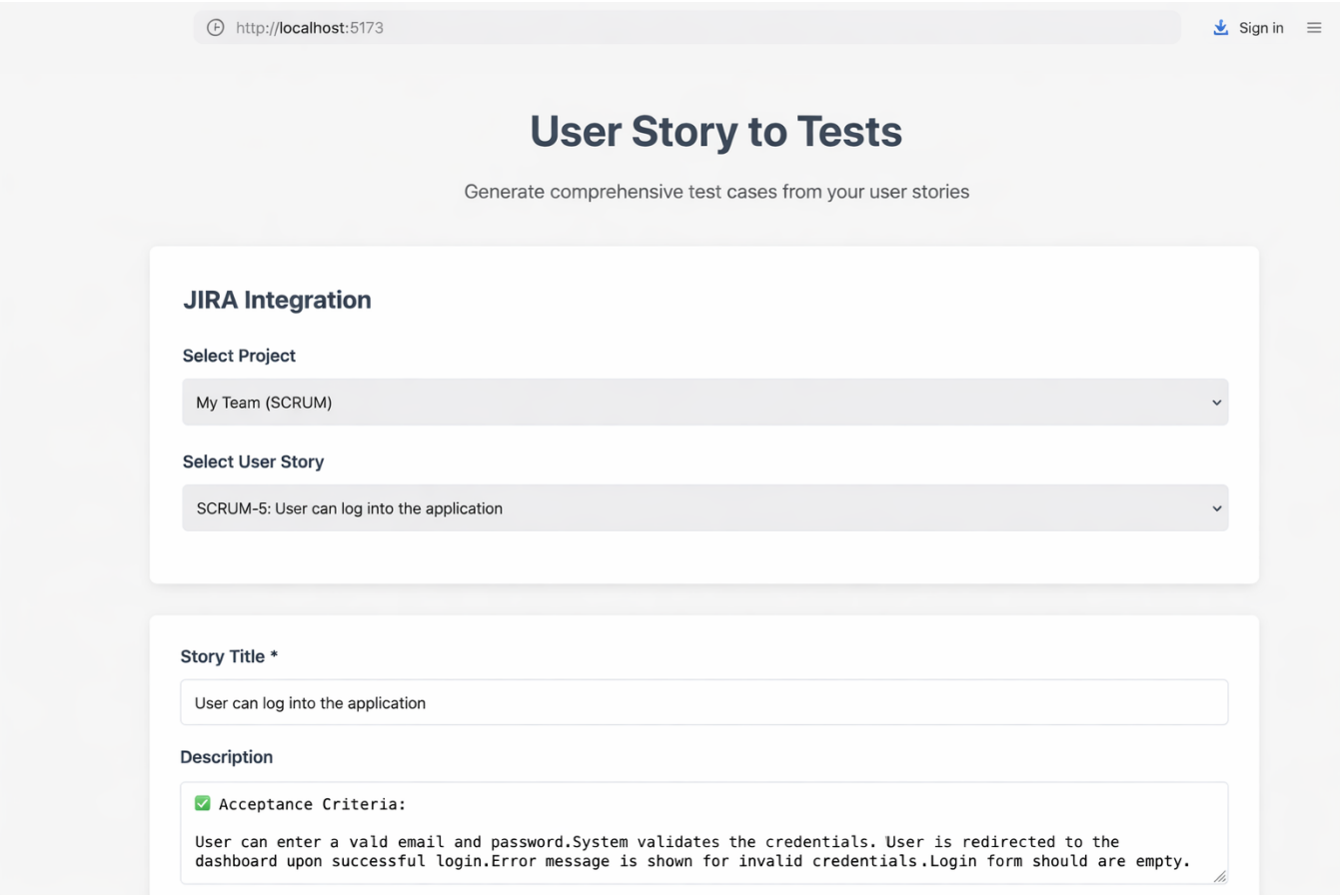
- **Postman Collection:** https://www.postman.com/prathibaviswanathan-1745800/...
- **Jira URL & Default Project:** https://prathibaviswanathan-genai.atlassian.net/jira/software/projects/GT/boards/2/backlog
- **Auth preference:** API Tokens

Confirm:

1. Start with Phase 1 (Backend Jira Client)? ✓ / ✗
2. Proceed sequentially through all phases after each approval? ✓ / ✗

Awaiting your approval to begin Phase 1.

Updated UI



http://localhost:5173

Generate

Generated Test Cases

20 test case(s) generated • Model: meta-llama/llama-4-cout-17{anser • Tokens: 2823

Test Case ID	Title	Category	Priority	Expected Result
TC-001	Valid Login Credentials	Positive	High	User is redirected to the dashboard
TC-002	Invalid Email	Negative	High	Error message is shown for invalid credentials
TC-003	Invalid Password	Negative	High	Error message is shown for invalid credentials
TC-004	Empty Email and Password	Negative	High	Login form does not submit
TC-005	Empty Email	Negative	High	Login form does not submit
TC-006	Empty Password	Negative	High	Login form does not submit
TC-007	Valid Email and Invalid Password	Negative	Medium	Error message is shown for invalid credentials